

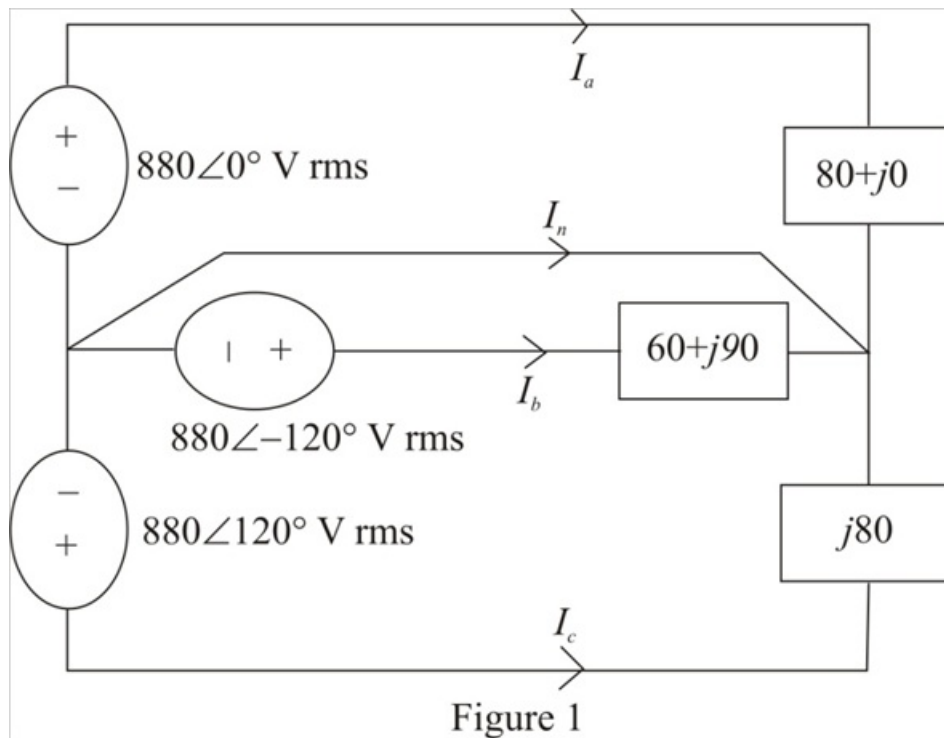
Problem

A balanced three-phase Y-source with $V_P = 880$ V rms drives a Y-connected three-phase load with phase impedance $\mathbf{Z}_A = 80 \Omega$, $\mathbf{Z}_B = 60 + j90 \Omega$, and $\mathbf{Z}_C = j80 \Omega$. Calculate the line currents and total complex power delivered to the load. Assume that the neutrals are connected.

Step-by-step solution

Step 1 of 5

Consider the circuit diagram shown in Figure 1. Assume the sources to be in positive sequence.



Step 2 of 5

Calculate the value of the current, I_a .

$$\begin{aligned} I_a &= \frac{880\angle 0^\circ}{80 + j0} \\ &= \frac{880\angle 0^\circ}{80\angle 0^\circ} \\ &= 11\angle 0^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_a is, $11\angle 0^\circ \text{ A rms}$.

Step 3 of 5

Calculate the value of the current, I_b .

$$\begin{aligned} I_b &= \frac{880\angle -120^\circ}{60 + j90} \\ &= \frac{880\angle -120^\circ}{108.167\angle 56.31^\circ} \\ &= 8.136\angle -176.31^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_b is, $8.136\angle -176.31^\circ \text{ A rms}$.

Step 4 of 5

Calculate the value of the current, I_c .

$$\begin{aligned} I_c &= \frac{880\angle 120^\circ}{j80} \\ &= \frac{880\angle 120^\circ}{80\angle 90^\circ} \\ &= 11\angle 30^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_c is, $11\angle 30^\circ \text{ A rms}$.

Step 5 of 5

Calculate the total complex power absorbed P_T by the load.

$$P_T = V_{an}I_a^* + V_{bn}I_b^* + V_{cn}I_c^*$$

Substitute $880\angle 0^\circ \text{ V rms}$ for V_{an} , $880\angle -120^\circ \text{ V rms}$ for V_{bn} , $880\angle 120^\circ \text{ V rms}$ for V_{cn} , $11\angle 0^\circ \text{ A rms}$ for I_a , $8.136\angle -176.31^\circ \text{ A rms}$ for I_b and $11\angle 30^\circ \text{ A rms}$ for I_c .

$$\begin{aligned} P_T &= (880\angle 0^\circ)(11\angle 0^\circ)^* + (880\angle -120^\circ)(8.136\angle -176.31^\circ)^* + (880\angle 120^\circ)(11\angle 30^\circ)^* \\ &= (880\angle 0^\circ)(11\angle 0^\circ) + (880\angle -120^\circ)(8.136\angle 176.31^\circ) + (880\angle 120^\circ)(11\angle -30^\circ) \\ &= 9680\angle 0^\circ + 7159.68\angle 56.31^\circ + 9680\angle 90^\circ \\ &= 20.76\angle 48.88^\circ \text{ kW} \end{aligned}$$

Therefore, the value of the total power absorbed P_T by the load is $20.76\angle 48.88^\circ \text{ kW}$.